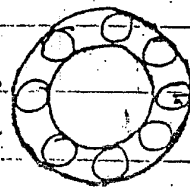


ENVELOPE

"The locus of the limiting position of the pts of X_n of any two curves of the family when one of them tends to coincide with the other which is kept fixed."



"A curve that is tangential to each of a family of curves"
Envelope of Normals to a curve is the Evolute of curve.

An annulus as the Envelope of the family of circles.

Ex 7.9

Working Rule.

curve: $f(x, y, z)$ — (1)Find $\frac{\partial f}{\partial z} = 0$ — (2)Eliminate z from (1) & (2)
we get Envelope.

QNo1 $y = mx + \frac{a}{m}$

$$f(x, y, m) = mx + \frac{a}{m} - y = 0 \quad (1)$$

$$\frac{\partial f}{\partial m} = x - \frac{a}{m^2} = 0$$

$$x - \frac{a}{m^2} = 0$$

$$x = \frac{a}{m^2}$$

$$m^2 = \frac{a}{x}$$

(2)

from (1) $mx + \frac{a}{m} - y = 0$

$$m^2 x + a - my = 0$$

LCM

$$\frac{a}{x} x + a - \sqrt{\frac{a}{x}} y = 0$$

$$2a = \sqrt{\frac{a}{x}} y$$

$$4a^2 = \frac{a}{x} y^2$$

$$\frac{4a^2 x}{a} = y^2$$

$$\Rightarrow y^2 = 4ax$$

7.9-2

Available at
www.mathcity.org

(b) No 2 (i) $f(x, y, z) = Ax^2 + Bx + C = 0$ — (1)

$$\frac{\partial f}{\partial x} = 2Ax + B = 0$$

$$2Ax + B = 0 \Rightarrow x = -\frac{B}{2A} \text{ — (ii)}$$

Putting (ii) in (1)

$$A\left(\frac{B^2}{4A^2}\right) + B\left(-\frac{B}{2A}\right) + C = 0$$

$$\frac{B^2}{4A} - \frac{B^2}{2A} + C = 0$$

$$B^2 - 2B^2 + 4AC = 0 \Rightarrow -B^2 + 4AC = 0$$

$$\Rightarrow B^2 - 4AC = 0$$

(b) $f(x, y, z) = Ax^3 + 3Bx^2 + 3Cx + D = 0$ — (1)

$$\frac{\partial f}{\partial x} = 3Ax^2 + 6Bx + 3C = 0$$

$$3Ax^2 + 6Bx + 3C = 0$$

$$Ax^2 + 2Bx + C = 0 \Rightarrow Ax^2 = -2Bx - C$$

from (1) $Ax^3 + 3Cx + 3Bx^2 + D = 0$

$$x(Ax^2 + 3C) + 3Bx^2 + D = 0$$

$$x(-2Bx - C + 3C) + 3Bx^2 + D = 0$$

$$-2Bx^2 + 2Cx + 3Bx^2 + D = 0$$

$$Bx^2 + 2Cx + D = 0 \text{ — (3)}$$

7.9-3

$$= \sec^2 \theta \left(x - \frac{g x^2}{u^2} \tan \theta \right) = 0$$

Available at
www.mathcity.org

$$x - \frac{g x^2}{u^2} \tan \theta = 0$$

$$\Rightarrow \frac{u^2 - g x^2 \tan \theta}{u^2} = 0 \Rightarrow u^2 - g x^2 \tan \theta = 0$$

$$\Rightarrow \frac{u^2}{g x^2} = \tan \theta \quad (2)$$

Eliminate θ from (1) & (2)

$$x \tan \theta - y - \frac{g x^2}{2u^2} \sec^2 \theta = 0 \quad \therefore \sec^2 \theta = 1 + \tan^2 \theta$$

$$x \left(\frac{u^2}{g x^2} \right) - y - \frac{g x^2}{2u^2} \left(1 + \frac{u^4}{g^2 x^2} \right) = 0$$

$$\frac{u^2}{g} - y - \frac{g x^2}{2u^2} - \frac{g x^2}{2u^2} \cdot \frac{u^4}{g^2 x^2} = 0$$

$$\frac{u^2}{g} - y - \frac{g x^2}{2u^2} - \frac{u^2}{2g} = 0$$

$$y = -\frac{g x^2}{2u^2} - \frac{u^2}{2g} + \frac{u^2}{g} = \frac{-g x^2 - u^4 + 2u^4}{2u^2 g}$$

$$y = \frac{u^4 - g x^2}{2u^2 g}$$

$$2u^2 g y = u^4 - g x^2 \text{ Ans.}$$

7.9-4

Solving (2) & (3)

$$A\alpha^2 + 2B\alpha + C = 0$$

$$B\alpha^2 + 2C\alpha + D = 0$$

$$\frac{\alpha^2}{2BD - 2C^2} = \frac{-\alpha}{AD - BC} = \frac{1}{2AC - 2B^2}$$

$$\alpha^2 = \frac{2BD - 2C^2}{2AC - 2B^2} = \frac{2(BD - C^2)}{2(AC - B^2)} \quad (4)$$

$$-\alpha = \frac{AD - BC}{2(AC - B^2)} \Rightarrow \alpha = \frac{BC - AD}{2(AC - B^2)} \quad (5)$$

Comparing (4) & (5)

$$\frac{BD - C^2}{AC - B^2} = \frac{(BC - AD)^2}{4(AC - B^2)^2}$$

$$(BD - C^2)4(AC - B^2) = (BC - AD)^2$$

$$(BC - AD)^2 = 4(BD - C^2)(AC - B^2)$$

Q3 $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$ (1)

$$f(x, \theta) = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} = x \tan \theta - \frac{gx^2 \sec^2 \theta}{2u^2} = 0$$

$$\frac{\partial f}{\partial \theta} = x \sec^2 \theta - 0 - \frac{gx^2}{2u^2} \sec \theta (\sec \theta \tan \theta) = 0$$

7.9-5

Q No 4

$$y = mx + \sqrt{a^2 m^2 + b^2}$$

$$f(x, y, m) = mx - y + \sqrt{a^2 m^2 + b^2} = 0 \quad \text{--- (1)}$$

$$\frac{\partial f}{\partial m} = x - 0 + \frac{1}{2} \frac{2a^2 m}{\sqrt{a^2 m^2 + b^2}} = 0$$

$$x = \frac{-a^2 m}{\sqrt{a^2 m^2 + b^2}} \Rightarrow \sqrt{a^2 m^2 + b^2} = \frac{-a^2 m}{x}$$

Squaring

$$a^2 m^2 + b^2 = \frac{a^4 m^2}{x^2}$$

$$b^2 = \frac{a^4 m^2}{x^2} - a^2 m^2 \Rightarrow b^2 = a^2 m^2 \left(\frac{a^2}{x^2} - 1 \right)$$

$$b^2 = a^2 m^2 \left(\frac{a^2 - x^2}{x^2} \right) \Rightarrow \frac{b^2 x^2}{a^2 - x^2} = a^2 m^2$$

$$\frac{b^2 x^2}{a^2 (a^2 - x^2)} = m^2 \quad \text{--- (2)}$$

Eliminating m from $f(x, y, m) = 0$ for Envelope using (2) in (1)

$$mx - y + \sqrt{a^2 m^2 + b^2} = 0$$

$$mx - y - \frac{a^2 m}{x} = 0$$

$$\frac{mx - a^2 m}{x} = y \Rightarrow y = m \left(\frac{x^2 - a^2}{x} \right)$$

$$\Rightarrow m = \frac{xy}{x^2 - a^2} = \frac{-xy}{a^2 - x^2} \Rightarrow m^2 = \frac{x^2 y^2}{(a^2 - x^2)^2} \quad \text{--- (3)}$$

Equating (2) & (3)

$$\frac{b^2 x^2}{a^2 (a^2 - x^2)} = \frac{x^2 y^2}{(a^2 - x^2)^2}$$

$$\frac{b^2}{a^2} = \frac{y^2}{(a^2 - x^2)}$$

$$\Rightarrow b^2 (a^2 - x^2) = a^2 y^2$$

$$\Rightarrow b^2 x^2 + a^2 y^2 = a^2 b^2$$

$$\Rightarrow \frac{b^2 x^2}{a^2} + \frac{a^2 y^2}{b^2} = a^2$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

7.9-6

Available at
www.mathcity.org

⑤ $(x-\alpha)^2 + y^2 = \alpha^2$

$f(x, y, \alpha) = (x-\alpha)^2 + y^2 - \alpha^2 = 0$

$\frac{\partial f}{\partial \alpha} = 2(x-\alpha)(-1) + 0 - 2\alpha = 0$

$-2x + 2\alpha - 2\alpha = 0$

$-2x = 0 \Rightarrow x = 0$

Elimination of α is not possible

OR α is absent in $\frac{\partial f}{\partial \alpha} = 0$ which is to be eliminated.

Hence No Envelope.

⑤ Let C be the centre of ellipse and CP & CD be its semi-conjugate diameters. If $P(a \cos \theta, b \sin \theta)$ & D is $(-a \sin \theta, b \cos \theta)$ then eq of PD is

$y - y_1 = m(x - x_1)$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$y - b \sin \theta = \frac{b(\sin \theta - \cos \theta)(x - a \cos \theta)}{a(\sin \theta + \cos \theta)}$

$= \frac{b \sin \theta - b \sin \theta}{-a \sin \theta - a \cos \theta}$
 $= \frac{b \sin \theta - b \cos \theta}{a(\sin \theta + \cos \theta)}$

$xy(\sin \theta + \cos \theta) - ab \sin \theta (\sin \theta + \cos \theta) = (b \sin \theta - b \cos \theta)(x - a \cos \theta)$

$xy \sin \theta + xy \cos \theta - ab \sin^2 \theta - ab \sin \theta \cos \theta = x b \sin \theta - ab \sin \theta \cos \theta - b x \cos \theta + ab \cos^2 \theta$

$xy (\sin \theta + \cos \theta) = b x (\sin \theta - \cos \theta) + ab (\cos^2 \theta + \sin^2 \theta)$

$b x (\sin \theta - \cos \theta) - xy (\sin \theta + \cos \theta) + ab = 0$

$\frac{b}{x} (\sin \theta - \cos \theta) - \frac{y}{b} (\sin \theta + \cos \theta) + 1 = 0$

①

7.9-7

Available at
www.mathcity.org

$$\frac{\partial f}{\partial \theta} = 0$$

$$\frac{x}{a} (\cos \theta + \sin \theta) - \frac{y}{b} (\cos \theta - \sin \theta) = 0. \quad (2)$$

Eliminating θ from (1) & (2)

$$\text{from (1)} \quad \frac{x}{a} (\sin \theta - \cos \theta) - \frac{y}{b} (\sin \theta + \cos \theta) = -1 \quad (3)$$

Squaring (2) & (3) and adding

$$\left[\frac{x}{a} (\cos \theta + \sin \theta) - \frac{y}{b} (\cos \theta - \sin \theta) \right]^2 + \left[\frac{x}{a} (\sin \theta - \cos \theta) - \frac{y}{b} (\sin \theta + \cos \theta) \right]^2 = (-1)^2$$

$$\Rightarrow \left[\frac{x^2}{a^2} (\cos \theta + \sin \theta)^2 + \frac{y^2}{b^2} (\cos \theta - \sin \theta)^2 - \frac{2xy}{ab} (\cos^2 \theta - \sin^2 \theta) \right] + \left[\frac{x^2}{a^2} (\sin \theta - \cos \theta)^2 + \frac{y^2}{b^2} (\sin \theta + \cos \theta)^2 - \frac{2xy}{ab} (\sin^2 \theta - \cos^2 \theta) \right] = 1$$

$$\Rightarrow \frac{x^2}{a^2} \left[(\cos \theta + \sin \theta)^2 + (\sin \theta - \cos \theta)^2 \right] + \frac{y^2}{b^2} \left[(\cos \theta - \sin \theta)^2 + (\sin \theta + \cos \theta)^2 \right] - \frac{2xy}{ab} (\cos^2 \theta - \sin^2 \theta) + \frac{2xy}{ab} (\cos^2 \theta - \sin^2 \theta) = 1$$

$$\Rightarrow \frac{x^2}{a^2} \left[(\cos^2 \theta + \sin^2 \theta) + 2 \sin \theta \cos \theta + (\sin^2 \theta + \cos^2 \theta) - 2 \sin \theta \cos \theta \right] + \frac{y^2}{b^2} \left[(\cos^2 \theta + \sin^2 \theta) - 2 \sin \theta \cos \theta + (\sin^2 \theta + \cos^2 \theta) + 2 \sin \theta \cos \theta \right] = 1$$

$$\Rightarrow \frac{x^2}{a^2} (1+1) + \frac{y^2}{b^2} (1+1) = 1$$

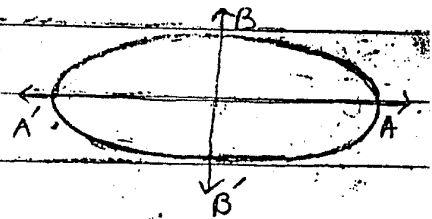
$$\frac{2x^2}{a^2} + \frac{2y^2}{b^2} = 1$$

Envelope

7.9-8

⑧ Given that $AA' + BB' = 2a$

Consider the ellipse



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \begin{matrix} AA' = 2a \\ BB' = 2b \end{matrix}$$

$$2a + 2b = 2a \quad (\text{from given})$$

$$a + b = a \Rightarrow b = a - a$$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ becomes } \frac{x^2}{a^2} + \frac{y^2}{(a-a)^2} = 1 \quad \left\{ \begin{array}{l} \text{One parameter} \\ \text{family with} \\ \text{parameter } a \\ \text{a is const} \end{array} \right.$$

$$f(x, y, a) = \frac{x^2}{a^2} + \frac{y^2}{(a-a)^2} - 1 = 0$$

$$\frac{\partial f}{\partial a} = -\frac{2x^2}{a^3} + \frac{2y^2(-1)}{(a-a)^3} = 0 \Rightarrow 2 \left(-\frac{x^2}{a^3} + \frac{y^2}{(a-a)^3} \right) = 0$$

$$\Rightarrow \frac{y^2}{(a-a)^3} = \frac{x^2}{a^3} \Rightarrow \frac{y^2}{x^2} = \frac{(a-a)^3}{a^3}$$

$$\Rightarrow \left(\frac{y}{x} \right) = \left(\frac{a-a}{a} \right)^3 \Rightarrow \left(\frac{y}{x} \right)^{2/3} = \left(\frac{a-a}{a} \right)^2$$

$$\Rightarrow \frac{y^{2/3}}{x^{2/3}} = \frac{a-a}{a} \Rightarrow 1 + \frac{y^{2/3}}{x^{2/3}} = \frac{a}{a}$$

$$\Rightarrow \frac{x^{2/3} + y^{2/3}}{x^{2/3}} = \frac{a}{a} \Rightarrow \frac{x^{2/3} + y^{2/3}}{a x^{2/3}} = 1$$

$$\Rightarrow a = \frac{x^{2/3} + y^{2/3}}{x^{2/3}}$$

Available at
www.mathcity.org

7.9-9

For envelope we eliminate α from ① & ②
from ①

$$\frac{x^2}{a^2} + \frac{y^2}{(a-x)^2} - 1 = 0$$

$$\frac{x^2}{\left(\frac{ax^{2/3}}{x^{2/3}+y^{2/3}}\right)^2} + \frac{y^2}{\left(a - \frac{ax^{2/3}}{x^{2/3}+y^{2/3}}\right)^2} = 1$$

$$\frac{x^2 \left(x^{2/3} + y^{2/3}\right)^2}{a^2 x^{4/3}} + \frac{y^2 \left(x^{2/3} + y^{2/3}\right)^2}{\left(ax^{2/3} + ay^{2/3} - ax\right)^2} = 1$$

$$\frac{x^{2-\frac{4}{3}} \left(x^{2/3} + y^{2/3}\right)^2}{a^2} + \frac{y^2 \left(x^{2/3} + y^{2/3}\right)^2}{a^2 y^{4/3}} = 1$$

$$\frac{x^{2/3} \left(x^{2/3} + y^{2/3}\right)^2}{a^2} + \frac{y^{2-\frac{4}{3}} \left(x^{2/3} + y^{2/3}\right)^2}{a^2} = 1$$

$$\left(x^{2/3} + y^{2/3}\right)^2 \left[\frac{x^{2/3}}{a^2} + \frac{y^{2/3}}{a^2}\right] = 1$$

$$\Rightarrow \left(x^{2/3} + y^{2/3}\right)^2 \left(x^{2/3} + y^{2/3}\right) = a^2$$

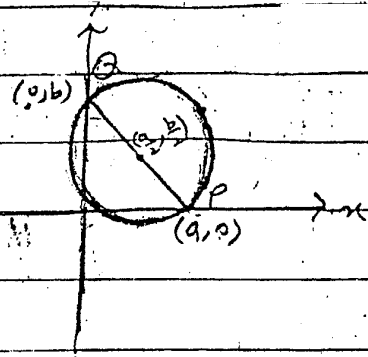
$$\Rightarrow \left(x^{2/3} + y^{2/3}\right)^3 = a^2$$

$$\Rightarrow x^{2/3} + y^{2/3} = a^{2/3} \text{ Ans}$$

Available at
www.mathcity.org

$$7.9 - 1c$$

⑦ Let the fixed st lines at right angles be the coord axes. Let $P(a, 0)$ & $Q(0, b)$ be the coord of the extremities of the sliding line. Centre of PQ is $(\frac{a}{2}, \frac{b}{2})$ and $|PQ| = c$ i radius = $\frac{c}{2}$.



$$\sqrt{(0-a)^2 + (b-0)^2} = c$$

Distance Formula

$$a^2 + b^2 = c^2$$

Squaring

Eq of circle with PQ as diameter is

$$(x - \frac{a}{2})^2 + (y - \frac{b}{2})^2 = (\frac{c}{2})^2$$

$$x^2 + \frac{a^2}{4} - 2x\frac{a}{2} + y^2 + \frac{b^2}{4} - 2y\frac{b}{2} = \frac{c^2}{4}$$

$$x^2 + y^2 - ax - by + \left(\frac{a^2 + b^2}{4}\right) = \frac{c^2}{4}$$

$$x^2 + y^2 - ax - by = \frac{c^2}{4} - \frac{a^2 + b^2}{4} = 0$$

$$\therefore c^2 = a^2 + b^2$$

$$x^2 + y^2 - ax - by = 0$$

$$\therefore c^2 = a^2 + b^2$$

$$c^2 - a^2 = b^2$$

a is parameter

c is const

$$f(x, y, a) = x^2 + y^2 - ax - \sqrt{c^2 - a^2} y = 0 \quad \text{--- ①}$$

$$\frac{\partial f}{\partial a} = -x - \frac{1}{2\sqrt{c^2 - a^2}} (-2a) y = 0$$

$$-x + \frac{ay}{\sqrt{c^2 - a^2}} = 0$$

$$\Rightarrow x = \frac{ay}{\sqrt{c^2 - a^2}}$$

$$\Rightarrow x^2 = \frac{a^2 y^2}{c^2 - a^2}$$

$$\Rightarrow \frac{1}{x^2} = \frac{c^2 - a^2}{a^2 y^2}$$

$$\Rightarrow \frac{1}{x^2} = \frac{c^2}{a^2 y^2} - \frac{1}{y^2}$$

$$\Rightarrow \frac{1}{x^2} + \frac{1}{y^2} = \frac{c^2}{a^2 y^2}$$

7.9 - 11

$$\frac{y^2 + x^2}{x^2 y^2} = \frac{c^2}{a^2 y^2} \Rightarrow a^2 = \frac{c^2 x^2 y^2}{y^2 (y^2 + x^2)}$$

$$\Rightarrow a^2 = \frac{c^2 x^2}{y^2 + x^2} \quad (2)$$

Eliminating a from envelope $y^2 + x^2$

from (1) $x^2 + y^2 - ax - \sqrt{c^2 - a^2} y = 0$

$$\frac{x^2 + y^2 - (cx)x}{\sqrt{y^2 + x^2}} - \frac{\sqrt{c^2 - c^2 x^2}}{x^2 + y^2} y = 0$$

$$\frac{x^2 + y^2 - cx^2}{\sqrt{y^2 + x^2}} - \frac{\sqrt{c^2 x^2 + c^2 y^2 - c^2 x^2}}{\sqrt{x^2 + y^2}} y = 0$$

$$\frac{(x^2 + y^2)(\sqrt{x^2 + y^2}) - cx^2 - (cy)y}{\sqrt{x^2 + y^2}} = 0$$

$$(x^2 + y^2)(\sqrt{x^2 + y^2}) - c(x^2 + y^2) = 0$$

$$(x^2 + y^2) [\sqrt{x^2 + y^2} - c] = 0$$

$$\sqrt{x^2 + y^2} - c = 0$$

$$\sqrt{x^2 + y^2} = c$$

$$x^2 + y^2 = c^2 \text{ Ans.}$$

7.9-12

(8) $\frac{x}{a} + \frac{y}{b} = 1$

(i) $a+b=c$ $b=c-a$

$\frac{x}{a} + \frac{y}{c-a} = 1$

one parameter family of lines

$f(x,y,a) = \frac{x}{a} + \frac{y}{c-a} - 1 = 0$

①

Available at
www.mathcity.org

$\frac{\partial f}{\partial a} = -\frac{x}{a^2} - \frac{y}{(c-a)^2} (-1) = 0$

$-\frac{x}{a^2} + \frac{y}{(c-a)^2} = 0$

$\Rightarrow \frac{x}{a^2} = \frac{y}{(c-a)^2}$

$\Rightarrow \frac{(c-a)^2}{a^2} = \frac{y}{x}$

$\Rightarrow \left(\frac{c-a}{a}\right)^2 = \frac{y}{x}$

$\Rightarrow \left(\frac{c}{a} - 1\right)^2 = \frac{y}{x}$

$\Rightarrow \frac{c}{a} - 1 = \frac{y^{\frac{1}{2}}}{x^{\frac{1}{2}}}$

$\frac{c}{a} = 1 + \frac{y^{\frac{1}{2}}}{x^{\frac{1}{2}}}$

$\Rightarrow \frac{c}{a} = \frac{x^{\frac{1}{2}} + y^{\frac{1}{2}}}{x^{\frac{1}{2}}}$

$\frac{cx^{\frac{1}{2}}}{x^{\frac{1}{2}} + y^{\frac{1}{2}}} = a$

②

Eliminating 'a' from ① & ② for Envelope

$\frac{x}{a} + \frac{y}{c-a} - 1 = 0$

$\Rightarrow \frac{x}{\frac{cx^{\frac{1}{2}}}{x^{\frac{1}{2}} + y^{\frac{1}{2}}}} + \frac{y}{c - \frac{cx^{\frac{1}{2}}}{x^{\frac{1}{2}} + y^{\frac{1}{2}}}} - 1 = 0$

$\Rightarrow \frac{(\sqrt{x} + \sqrt{y})x}{c\sqrt{x}} + \frac{y(\sqrt{x} + \sqrt{y})}{c\sqrt{x} + c\sqrt{y} - c\sqrt{x}} - 1 = 0$

7.9-14

$$x = \frac{y a^{\frac{n-1}{2}}}{(c^n - a^n)^{\frac{n+1}{n}}} \Rightarrow \frac{x}{y} = \frac{a^{\frac{n+1}{2}}}{(c^n - a^n)^{\frac{n+1}{n}}}$$

Taking power $\frac{n}{n+1}$ on both sides

$$\left(\frac{x}{y}\right)^{\frac{n}{n+1}} = \left(\frac{a^{\frac{n+1}{2}}}{(c^n - a^n)^{\frac{n+1}{n}}}\right)^{\frac{n}{n+1}}$$

$$\frac{x^{\frac{n}{n+1}}}{y^{\frac{n}{n+1}}} = \frac{a^n}{c^n - a^n} \Rightarrow \frac{y^{\frac{n}{n+1}}}{x^{\frac{n}{n+1}}} = \frac{c^n - a^n}{a^n}$$

$$\frac{y^{\frac{n}{n+1}}}{x^{\frac{n}{n+1}}} = \frac{c^n}{a^n} - 1 \Rightarrow \frac{1 + y^{\frac{n}{n+1}}}{x^{\frac{n}{n+1}}} = \frac{c^n}{a^n}$$

$$\frac{y^{\frac{n}{n+1}} + 1}{x^{\frac{n}{n+1}}} = \frac{c^n}{a^n} \Rightarrow a = \frac{c x^{\frac{n}{n+1}}}{x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}}}$$

$$\left[a\right]^{\frac{1}{n}} = \left[\frac{c x^{\frac{n}{n+1}}}{x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}}}\right]^{\frac{1}{n}} \Rightarrow a = \frac{c x^{\frac{1}{n+1}}}{\left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}}\right)^{\frac{1}{n}}} \quad (2)$$

Eliminating a from (1) using (2)

$$\frac{x}{a} + \frac{y}{(c^n - a^n)^{\frac{1}{n}}} - 1 = 0$$

$$\frac{x}{c x^{\frac{1}{n+1}}} + \frac{y}{\left(c^n - \left(\frac{c x^{\frac{1}{n+1}}}{\left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}}\right)^{\frac{1}{n}}}\right)^n\right)^{\frac{1}{n}}} - 1 = 0$$

7.9-15

$$\frac{x \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{c x^{\frac{1}{n+1}}} + \frac{y \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{\left[c^{\frac{n}{n+1}} x + c y^{\frac{n}{n+1}} - c x^{\frac{n}{n+1}} \right]^{\frac{1}{n}}} - 1 = 0$$

$$\therefore \frac{x \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{c x^{\frac{1}{n+1}}} + \frac{y \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{\left[c y^{\frac{n}{n+1}} \right]^{\frac{1}{n}}} - 1 = 0$$

$$\ominus \quad \frac{x^{1-\frac{1}{n+1}} \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{c} + \frac{y \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{c y^{\frac{1}{n+1}}} - 1 = 0$$

$$\frac{x^{\frac{n}{n+1}} \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{c} + \frac{y^{1-\frac{1}{n+1}} \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{c} - 1 = 0$$

(iii)

$$\frac{x^{\frac{n}{n+1}} \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{x} + \frac{y^{\frac{n}{n+1}} \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{y} - c = 0 \quad \text{By LCM}$$

$$\ominus \quad \frac{\left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{1}{n}}}{\left[x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right]} - c = 0$$

$$\left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{1+\frac{1}{n}} - c = 0 \quad \Rightarrow \quad \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right)^{\frac{n+1}{n}} = c$$

Taking $\frac{n}{n+1}$ power on both sides

$$\left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} \right) = c^{\frac{n}{n+1}} \quad \text{Ans.}$$

7.9-16

29

Q9 $x = a(\cos t + t \sin t)$

$y = a(\sin t - t \cos t)$

$\frac{dx}{dt} = a(-\sin t + \sin t + t \cos t)$

$\frac{dy}{dt} = a(\cos t) - \cos t - t(-\sin t)$

$\frac{dx}{dt} = a t \cos t$

$\frac{dy}{dt} = a t \sin t$

$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$

$\frac{dy}{dx} = \frac{a t \sin t}{a t \cos t}$

$= \frac{\sin t}{\cos t} = m$

Available at
www.mathcity.org

Slope of the normal $= -\frac{1}{m} = -\frac{\cos t}{\sin t}$

Eq of Normal

$y - y_1 = -\frac{1}{m} (x - x_1)$

$y - a(\sin t - t \cos t) = -\frac{\cos t}{\sin t} (x - a(\cos t + t \sin t))$

$y \sin t - a \sin t (\sin t - t \cos t) = -x \cos t + a \cos t (\cos t + t \sin t)$

$x \cos t + y \sin t = a \sin t (\sin t - t \cos t) + a \cos t (\cos t + t \sin t)$

$x \cos t + y \sin t = a \sin^2 t - a t \sin t \cos t + a \cos^2 t + a \cos t t \sin t$

$x \cos t + y \sin t = a (\sin^2 t + \cos^2 t)$

$x \cos t + y \sin t = a$ (1)

$f(x, y, t) = x \cos t + y \sin t - a = 0$

$\frac{\partial f}{\partial t} = x(-\sin t) + y \cos t = 0$

$$-x \sin t + y \cos t = 0 \quad \text{--- 7.9 - 17 ---} \quad (2)$$

Squaring ① & ② & adding

$$(x \cos t + y \sin t)^2 + (-x \sin t + y \cos t)^2 = a^2$$

$$x^2 \cos^2 t + y^2 \sin^2 t + 2xy \cos t \sin t + x^2 \sin^2 t + y^2 \cos^2 t - 2xy \sin t \cos t = a^2$$

$$x^2 (\cos^2 t + \sin^2 t) + y^2 (\sin^2 t + \cos^2 t) = a^2$$

$$x^2 + y^2 = a^2 \quad \text{Required Evolute}$$

Available at

www.mathcity.org

Q10. Parametric eqs of astroid

$$x = a \cos^3 \theta$$

$$\frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta)$$

$$y = a \sin^3 \theta$$

$$\frac{dy}{d\theta} = 3a \sin^2 \theta (\cos \theta)$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{-3a \sin^2 \theta \cos \theta}{3a \cos^3 \theta \sin \theta} = -\tan \theta$$

$$\text{Slope of Normal} = -\frac{1}{dy/dx} = \cot \theta$$

Suppose normal makes angle ϕ with x-axis then slope of Normal is $\tan \phi$

$$\therefore \tan \phi = \cot \theta$$

$$\tan \phi = \tan \left(\frac{\pi}{2} - \theta \right) \Rightarrow \phi = \left(\frac{\pi}{2} - \theta \right)$$

$$\therefore x = a \cos^3 \theta = a \cos^3 \left(\frac{\pi}{2} - \phi \right) = a \sin^3 \phi$$

$$y = a \sin^3 \theta = a \sin^3 \left(\frac{\pi}{2} - \phi \right) = a \cos^3 \phi$$

$$\text{Eq of Normal } y - a \cos^3 \phi = \tan \phi (x - a \sin^3 \phi)$$

$$[x] + [y]$$

7.9-18

$$y - a \cos^3 \phi = \frac{\sin \phi}{\cos \phi} (x - a \sin^3 \phi)$$

$$y \cos \phi - a \cos^4 \phi = x \sin \phi - a \sin^4 \phi$$

$$x \sin \phi - y \cos \phi + a \cos^4 \phi - a \sin^4 \phi = 0$$

$$x \sin \phi - y \cos \phi + a (\cos^4 \phi - \sin^4 \phi) = 0$$

$$x \sin \phi - y \cos \phi + a (\cos^2 \phi - \sin^2 \phi) (\cos^2 \phi + \sin^2 \phi) = 0$$

$$x \sin \phi - y \cos \phi + a \cos 2\phi = 0 \quad \text{--- (1)}$$

$$f(x, y, \phi) = x \sin \phi - y \cos \phi + a \cos 2\phi = 0$$

$$\frac{\partial f}{\partial \phi} = x \cos \phi - y (-\sin \phi) + a (-\sin 2\phi)(2) = 0$$

$$x \cos \phi + y \sin \phi - 2a \sin 2\phi = 0 \quad \text{--- (2)}$$

Eliminate ϕ from (1) and (2) for Envelope Normal (which is Evolution of Astroid)

$$x \sin \phi - y \cos \phi + a \cos 2\phi = 0$$

$$x \cos \phi + y \sin \phi - 2a \sin 2\phi = 0$$

$$\frac{x}{2a \sin 2\phi \cos \phi - a \cos 2\phi \sin \phi} = \frac{-y}{-2a \sin 2\phi \sin \phi - \cos \phi a \cos 2\phi} = \frac{1}{\sin^2 \phi + \cos^2 \phi}$$

$$\frac{x}{2a \sin 2\phi \cos \phi - a \cos 2\phi \sin \phi} = \frac{y}{2a \sin 2\phi \sin \phi + \cos \phi a \cos 2\phi} = 1$$

$$x = 2a \sin 2\phi \cos \phi - a \cos 2\phi \sin \phi \quad \text{--- (3)}$$

$$y = 2a \sin 2\phi \sin \phi + a \cos 2\phi \cos \phi \quad \text{--- (4)}$$

7.9-20

$$= a(3 \sin \theta \cos \theta (\cos \theta + \sin \theta) + \cos^3 \theta + \sin^3 \theta)$$

$$x+y = a(\cos \theta + \sin \theta)^3$$

$$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

from (5) & (6)

$$x-y = 3a \sin \theta - 2a \sin^3 \theta - 3a \cos \theta + 2a \cos^3 \theta$$

$$= 3a \sin \theta - 2a \sin^3 \theta - (a \sin^3 \theta) + (a \sin^3 \theta) - 3a \cos \theta + 2a \cos^3 \theta + (a \cos^3 \theta) - (a \cos^3 \theta)$$

$$= 3a \sin \theta - 2a \sin^3 \theta + a \sin^3 \theta - 3a \cos \theta + 3a \cos^3 \theta - a \cos^3 \theta$$

$$= 3a \sin \theta (1 - \sin^2 \theta) + a \sin^3 \theta - 3a \cos \theta (1 - \cos^2 \theta) - a \cos^3 \theta$$

$$= 3a \sin \theta \cos^2 \theta + a \sin^3 \theta - 3a \cos \theta (\sin^2 \theta) - a \cos^3 \theta$$

$$= 3a \sin \theta \cos^2 \theta (-\cos \theta - \sin \theta) + a \sin^3 \theta - a \cos^3 \theta$$

$$= a [\sin^3 \theta - \cos^3 \theta - 3a \sin \theta \cos \theta (\sin \theta - \cos \theta)]$$

$$x-y = a(\sin \theta - \cos \theta)^3$$

Taking power $\frac{2}{3}$ and adding (7) & (8)

$$(x+y)^{\frac{2}{3}} + (x-y)^{\frac{2}{3}} = a^{\frac{2}{3}} \left[(\cos \theta + \sin \theta)^3 \right]^{\frac{2}{3}} + a^{\frac{2}{3}} \left[(\sin \theta - \cos \theta)^3 \right]^{\frac{2}{3}}$$

$$= a^{\frac{2}{3}} \left[(\cos^2 \theta + \sin^2 \theta)^2 + (\sin^2 \theta - \cos^2 \theta)^2 \right]$$

$$= a^{\frac{2}{3}} \left[\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta + \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta \right]$$

$$(x+y)^{\frac{2}{3}} + (x-y)^{\frac{2}{3}} = 2a^{\frac{2}{3}} \text{ Ans.}$$